STA 2002, Summer 2024
Probability and Statistics II

Analysis of Variance (ANOVA)

Example

A manufacturer of paper used for making grocery bags is interested in improving the tensile strength of the paper. They think that tensile strength is a function of the hardwood concentration in the pulp. Their team decides to investigate four levels of hardwood concentration: 5%, 10%, 15%, and 20%. They decide to make up six test specimens at each concentration level. All 24 specimens are tested on a laboratory tensile tester, in random order.

Table 13-1 Tensile Strength of Paper (psi)

Hardwood Concentration (%)	Observations						Totals	Averages
	1	2	3	4	5	6		
5	7	8	15	11	9	10	60	10.00
10	12	17	13	18	19	15	94	15.67
15	14	18	19	17	16	18	102	17.00
20	19	25	22	23	18	20	127	21.17
							383	15.96



- This is a completely randomized single-factor experiment with four levels of the factor
- The levels of the factor are sometimes called **treatments**, and each treatment has six observations or replicates

One-factor ANOVA

- *m* treatments, *i*-th treatment result in observations sampled from a distribution $N(\mu_i, \sigma^2)$
- n_i samples from the *i*-th treatment

$$X_{ij} \overset{i.i.d.}{\sim} N(\mu_i, \sigma^2), \quad j = 1, 2, \ldots, n_i. \qquad i = 1, \ldots, m$$

grand total $n = \sum_{i=1}^{m} n_i$

- Test the equality of the *m* means

$$H_0 : \mu_1 = \mu_2 = \ldots = \mu_m = \mu, \quad H_1 : H_0 \text{ is not true}$$

Table 9.3-1 One-factor random samples

					Means
X_1:	X_{11}	X_{12}	$\cdots$	X_{1n_1}	$\overline{X}_{1\cdot}$
X_2:	X_{21}	X_{22}	$\cdots$	X_{2n_2}	$\overline{X}_{2\cdot}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
X_m:	X_{m1}	X_{m2}	$\cdots$	X_{mn_m}	$\overline{X}_{m\cdot}$
Grand Mean:					$\overline{X}_{\cdot\cdot}$

Reading: Chapter 9.3

Notations

- Sample mean of the ith distribution:(treatments)

$$\boxed{\overline{X}_{i\cdot} = \frac{1}{n_i}\sum_{j=1}^{n_i} X_{ij}} \quad i = 1, 2, \ldots, m$$

- Overall sample mean (aka **grand mean**) of the entire data table:

$$\boxed{\overline{X}_{\cdot\cdot} = \frac{1}{n}\sum_{i=1}^{m}\sum_{j=1}^{n_i} X_{ij}} = \frac{1}{n}\sum_{i=1}^{m} n_i \overline{X}_{i\cdot}$$

- Overall sum-of-squares or **total sum-of-squares** $SS(TO)$:

$$\boxed{SS(TO) = \sum_{i=1}^{m}\sum_{j=1}^{n_i} \left(X_{ij} - \overline{X}_{\cdot\cdot}\right)^2}$$

Notations

- **Between-treatment sum-of-squares** $SS(T)$:

$$\boxed{SS(T) = \sum_{i=1}^{m} n_i \left(\overline{X}_{i\cdot} - \overline{X}_{\cdot\cdot}\right)^2}$$

- **Error sum-of-squares** or within-treatment sum-of-squares $SS(E)$:

$$\boxed{SS(E) = \sum_{i=1}^{m} \sum_{j=1}^{n_i} \left(X_{ij} - \overline{X}_{i\cdot}\right)^2}$$

- Within-treatment sum-of-squares of treatment i for $i = 1, 2, \ldots, m$, denoted by $SS(E_i)$:

$$SS(E_i) = \sum_{j=1}^{n_i} \left(X_{ij} - \overline{X}_{i\cdot}\right)^2$$

$$SS(E) = \sum_{i=1}^{m} SS(E_i)$$

Relations

$$\mathrm{SS(TO)} = \sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij} - \overline{X}_{..})^2$$

$$= \sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij} - \overline{X}_{i.} + \overline{X}_{i.} - \overline{X}_{..})^2$$

$$= \sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij} - \overline{X}_{i.})^2 + \sum_{i=1}^{m}\sum_{j=1}^{n_i}(\overline{X}_{i.} - \overline{X}_{..})^2$$

$$+ 2\sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij} - \overline{X}_{i.})(\overline{X}_{i.} - \overline{X}_{..}). \quad \text{=0}$$

$$= \sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij} - \overline{X}_{i.})^2 + \sum_{i=1}^{m} n_i(\overline{X}_{i.} - \overline{X}_{..})^2$$

$$\mathrm{SS(TO)} = \mathrm{SS(E)} + \mathrm{SS(T)}.$$

Hypothesis Testing

$$H_0 : \mu_1 = \mu_2 = \ldots = \mu_m = \mu, \quad H_1 : H_0 \text{ is not true}$$

Under H_0:

$$X_{i,j} \sim N(\mu, \sigma^2), \quad i = 1, \ldots, m, j = 1, \ldots, n_j$$

grand total $n = \sum_{i=1}^{m} n_i$

$$\Rightarrow \begin{cases} \dfrac{\mathrm{SS(TO)}}{\sigma^2} \sim \chi^2(n-1) \\ \dfrac{\mathrm{SS(E_i)}}{\sigma^2} \sim \chi^2(n_i - 1), i = 1, \ldots, m \end{cases} \Rightarrow \frac{\mathrm{SS(E)}}{\sigma^2} \sim \chi^2\left(\sum_{i=1}^{m}(n_i - 1)\right) = \chi^2(n-m)$$

Recall

$$\mathrm{SS(TO)} = \mathrm{SS(E)} + \mathrm{SS(T)}.$$

Theorem 9.3-1 Let $Q = Q_1 + Q_2 + \cdots + Q_k$, where $Q, Q_1, \ldots, Q_k$ are $k+1$ real quadratic forms in n mutually independent random variables normally distributed with the same variance σ^2. Let $Q/\sigma^2, Q_1/\sigma^2, \ldots, Q_{k-1}/\sigma^2$ have chi-square distributions with $r, r_1, \ldots, r_{k-1}$ degrees of freedom, respectively. If Q_k is nonnegative, then

(a) $Q_1, \ldots, Q_k$ are mutually independent, and hence,

(b) Q_k/σ^2 has a chi-square distribution with $r - (r_1 + \cdots + r_{k-1}) = r_k$ degrees of freedom.

$$\frac{\mathrm{SS(T)}}{\sigma^2} \sim \chi^2(m-1)$$

SS(E) and SS(T) are independent

Hypothesis Testing

Hypothesis

$$H_0 : \mu_1 = \mu_2 = \ldots = \mu_m = \mu, \quad H_1 : H_0 \text{ is not true}$$

Test statistics

$$F = \frac{\mathrm{SS(T)}/(m-1)}{\mathrm{SS(E)}/(n-m)} = \frac{[\mathrm{SS(T)}/\sigma^2]/(m-1)}{[\mathrm{SS(E)}/\sigma^2]/(n-m)}$$

Under H_0:

$$F \sim F(m-1, n-m)$$

$$\mathrm{SS(TO)} = \mathrm{SS(E)} + \mathrm{SS(T)}.$$

$\chi^2(n-1)$ $\chi^2(n-m)$ $\chi^2(m-1)$

SS(E) and SS(T) are independent

Critical (rejection) region:

$$F \geq F_\alpha(m-1, n-m).$$

***p*-value:**

$$P\{F(m-1, n-m) > F\}$$

One-factor ANOVA Table

Table 9.3-2 Analysis-of-variance table

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F Ratio
Treatment	SS(T)	$m-1$	$\mathrm{MS(T)} = \dfrac{\mathrm{SS(T)}}{m-1}$	$\dfrac{\mathrm{MS(T)}}{\mathrm{MS(E)}}$
Error	SS(E)	$n-m$	$\mathrm{MS(E)} = \dfrac{\mathrm{SS(E)}}{n-m}$	
Total	SS(TO)	$n-1$		

Example

Table 9.3-3 Illustrative data

	Observations			$\overline{X}_{i\cdot}$
x_1:	13	8	9	10
x_2:	15	11	13	13
x_3:	8	12	7	9
x_4:	11	15	10	12
$\overline{x}_{\cdot\cdot}$				11

single-factor experiment with 4 levels (treatments) of the factor, each treatment has 3 observations

$$m = 4, n_1 = n_2 = n_3 = n_4 = 3, n = 12$$

$$\mathrm{SS(TO)} = (13-11)^2 + (8-11)^2 + \cdots + (15-11)^2 + (10-11)^2 = 80,$$

$$\mathrm{SS(E)} = (13-10)^2 + (8-10)^2 + \cdots + (15-12)^2 + (10-12)^2 = 50,$$

$$\mathrm{SS(T)} = 3[(10-11)^2 + (13-11)^2 + (9-11)^2 + (12-11)^2] = 30.$$

$$\frac{\mathrm{SS(T)}/(m-1)}{\mathrm{SS(E)}/(n-m)} = \frac{30/3}{50/8} = 1.6 < 4.07 = F_{0.05}(3,8)$$

Fail to reject H_0

Table 9.3-4 ANOVA table for illustrative data

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	*F* Ratio	*p*-value
Treatment	30	3	30/3	1.6	0.264
Error	50	8	50/8		
Total	80	11			

Example

Table 9.3-5 Pullout test data

	Observations							$\sum_{j=1}^{7} x_{ij}$	$\sum_{j=1}^{7} x_{ij}^2$
x_1:	92	90	87	105	86	83	102	645	59,847
x_2:	100	108	98	110	114	97	94	721	74,609
x_3:	143	149	138	136	139	120	145	970	134,936
x_4:	147	144	160	149	152	131	134	1017	148,367
x_5:	142	155	119	134	133	146	152	981	138,415
Totals								4334	556,174

$$\mathrm{SS(TO)} = 556,174 - \frac{1}{35}(4334)^2 = 19,500.97$$

$$\mathrm{SS(T)} = \frac{1}{7}\left(645^2 + 721^2 + 970^2 + 1017^2 + 981^2\right) - \frac{1}{35}(4334)^2 = 16,672.11$$

$$\mathrm{SS(E)} = 19,500.97 - 16,672.11 = 2828.86$$

Example

Table 9.3-6 ANOVA table for pullout tests

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F
Treatment	16,672.11	4	4, 168.03	44.20
Error	2,828.86	30	94.30	
Total	19,500.97	34		

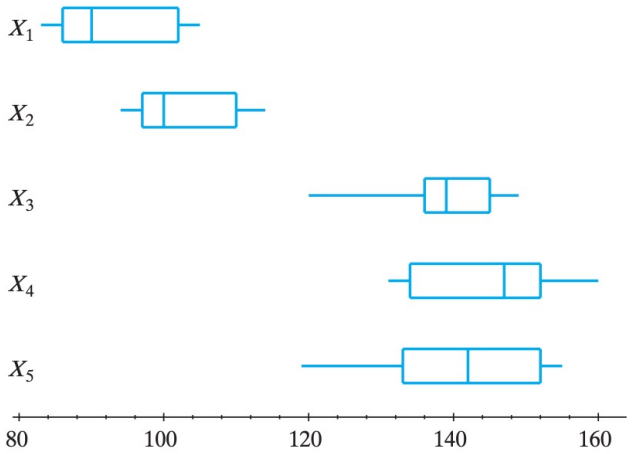


Figure 9.3-1 Box plots for pullout tests

$$F = \frac{16{,}672.11/4}{2828.86/30} = 44.20 \geq 4.02 = F_{0.01}(4, 30)$$

Reject H_0 at significance level 1%

Two-factor ANOVA

Reading: Chapter 9.4

Example

Each of three cars is driven with each of four different brands of gasoline. The number of miles per gallon (MPG) driven for each of the 3*4=12 different combinations is recorded in the following table.

Factor 2
(4 levels)

Factor 1
(3 levels)

	Gasoline				
Car	1	2	3	4	$\overline{X}_{i\cdot}$
1	26	28	31	31	29
2	24	25	28	27	26
3	25	25	28	26	26
$\overline{X}_{\cdot j}$	25	26	29	28	27

Two-factor ANOVA

- Two factors (attributes), one of which has a levels and the other b levels; There are thus n = ab possible combinations
- For the (i,j)-th combination, data are sampled from a distribution

$$X_{i,j} \sim N(\mu_{i,j}, \sigma^2)$$

- **One** samples from the (i,j)-th combination of treatment
- Assume

$$\mu_{ij} = \mu + \alpha_i + \beta_j$$

i-th row effect j-th column effect

Remark: without loss of generality, we assume $\sum_{i=1}^{a} \alpha_i = 0$ and $\sum_{j=1}^{b} \beta_j = 0$

Two-factor Hypothesis Testing

We want to test:

$$\mu_{ij} = \mu + \alpha_i + \beta_j$$

i-th row effect

j-th column effect

1. (Is there an effect on μ due to row?)

$$H_{0A} : \alpha_1 = \alpha_2 = \ldots = \alpha_a = 0, \quad H_{1A} : H_{0A} \text{ is not true}$$

2. (Is there an effect on μ due to column?)

$$H_{0B} : \beta_1 = \beta_2 = \ldots = \beta_b = 0, \quad H_{1B} : H_{0B} \text{ is not true}$$

Notations

- Sample mean of the i-th row factor

$$\overline{X}_{i\cdot} = \frac{1}{b}\sum_{j=1}^{b} X_{ij}$$

- Sample mean of the j-th column factor

$$\overline{X}_{\cdot j} = \frac{1}{a}\sum_{i=1}^{a} X_{ij}$$

- Overall sample mean of the entire data table

$$\overline{X}_{\cdot\cdot} = \frac{1}{ab}\sum_{i=1}^{a}\sum_{j=1}^{b} X_{ij}$$

Notations

- Overall sum-of-squares or total sum-of-squares

$$SS(TO) = \sum_{i=1}^{a} \sum_{j=1}^{b} \left(X_{ij} - \overline{X}_{..}\right)^2$$

- Sum-of-squares of Factor A (row factor)

$$SS(A) = b \sum_{i=1}^{a} \left(\overline{X}_{i.} - \overline{X}_{..}\right)^2$$

- Sum-of-squares of Factor B (column factor)

$$SS(B) = a \sum_{j=1}^{b} \left(\overline{X}_{.j} - \overline{X}_{..}\right)^2$$

Notations

- Error sum-of-squares

$$SS(E) = \sum_{i=1}^{a} \sum_{j=1}^{b} \left[X_{ij} - \underbrace{(\overline{X}_{i\cdot} - \overline{X}_{\cdot\cdot})}_{\widehat{\alpha}_i} - \underbrace{(\overline{X}_{\cdot j} - \overline{X}_{\cdot\cdot})}_{\widehat{\beta}_j} - \underbrace{\overline{X}_{\cdot\cdot}}_{\widehat{\mu}} \right]^2$$

- Relations

$$SS(TO) = SS(A) + SS(B) + SS(E)$$

Hypothesis Testing

$H_{0A}: \alpha_1 = \alpha_2 = \ldots = \alpha_a = 0, \quad H_{1A}: H_{0A}$ is not true

Under H_{0A}:

$$SS(A) = b\sum_{i=1}^{a}(\overline{X}_{i\cdot} - \overline{X}_{\cdot\cdot})^2$$

$$\frac{\mathrm{SS(A)}}{\sigma^2} \sim \chi^2(a-1)$$

SS(E) ⊥ SS(A)

$H_{0B}: \beta_1 = \beta_2 = \ldots = \beta_b = 0, \quad H_{1B}: H_{0B}$ is not true

Under H_{0B}:

$$SS(B) = a\sum_{j=1}^{b}(\overline{X}_{\cdot j} - \overline{X}_{\cdot\cdot})^2$$

$$\frac{\mathrm{SS(B)}}{\sigma^2} \sim \chi^2(b-1)$$

SS(E) ⊥ SS(B)

Further note that, $\frac{\mathrm{SS(E)}}{\sigma^2} \sim \chi^2\big((a-1)(b-1)\big)$

Then, F-test can be adopted to test H_{0A} or H_{0B}.

Hypothesis Testing

Hypothesis

$H_{0A}: \alpha_1 = \alpha_2 = \ldots = \alpha_a = 0, \quad H_{1A}: H_{0A}$ is not true

Test statistics

$$F_A = \frac{\mathrm{SS(A)}/[\sigma^2(a-1)]}{\mathrm{SS(E)}/[\sigma^2(a-1)(b-1)]} = \frac{\mathrm{SS(A)}/(a-1)}{\mathrm{SS(E)}/[(a-1)(b-1)]}$$

Under H_{0A}:

$$F_A \sim F(a-1, (a-1)(b-1))$$

Critical (rejection) region:

$$F_A \geq F_\alpha(a-1, (a-1)(b-1))$$

***p*-value:**

$$P\{F\ \ (a-1, (a-1)(b-1)) \geq F_A\}$$

Hypothesis

$H_{0B}: \beta_1 = \beta_2 = \ldots = \beta_b = 0, \quad H_{1B}: H_{0B}$ is not true

Test statistics

$$F_B = \frac{\mathrm{SS(B)}/[\sigma^2(b-1)]}{\mathrm{SS(E)}/[\sigma^2(a-1)(b-1)]} = \frac{\mathrm{SS(B)}/(b-1)}{\mathrm{SS(E)}/[(a-1)(b-1)]}$$

Under H_{0A}:

$$F_B \sim F(b-1, (a-1)(b-1))$$

Critical (rejection) region:

$$F_B \geq F_\alpha(b-1, (a-1)(b-1))$$

***p*-value:**

$$P\{F\ \ (b-1, (a-1)(b-1)) \geq F_B\}$$

Two-factor ANOVA Table

Table 9.4-1 Two-way ANOVA table, one observation per cell

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F
Factor A (row)	SS(A)	$a-1$	$\mathrm{MS(A)} = \dfrac{\mathrm{SS(A)}}{a-1}$	$\dfrac{\mathrm{MS(A)}}{\mathrm{MS(E)}}$
Factor B (column)	SS(B)	$b-1$	$\mathrm{MS(B)} = \dfrac{\mathrm{SS(B)}}{b-1}$	$\dfrac{\mathrm{MS(B)}}{\mathrm{MS(E)}}$
Error	SS(E)	$(a-1)(b-1)$	$\mathrm{MS(E)} = \dfrac{\mathrm{SS(E)}}{(a-1)(b-1)}$	
Total	SS(TO)	$ab-1$		

Example

Table 9.4-2 Gas mileage data

Car	Gasoline 1	2	3	4	$\overline{X}_{i\cdot}$
1	26	28	31	31	29
2	24	25	28	27	26
3	25	25	28	26	26
$\overline{X}_{\cdot j}$	25	26	29	28	27

Table 9.4-3 ANOVA table for gas mileage data

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F	p-value
Row (A)	24	2	12	18	0.003
Column (B)	30	3	10	15	0.003
Error	4	6	2/3		
Total	58	11			

$$\text{SS(B)} = 3[(25-27)^2 + (26-27)^2 + (29-27)^2 + (28-27)^2] = 30;$$

$$\text{SS(E)} = (26-29-25+27)^2 + (24-26-25+27)^2 + \cdots + (26-26-28+27)^2 = 4.$$

$$H_{0B} : \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0$$

$$\frac{\text{SS(B)}/(4-1)}{\text{SS(E)}/[(3-1)(4-1)]} = 15 > 9.78 = F_{0.01}(3,6).$$

Reject H_{0B} at significance level 1%

Two-factor ANOVA with c>1 observations per cell

- Two factors (attributes), one of which has a levels and the other b levels; There are thus n = ab possible interactions
- For the (i,j)-th combination, we have c observations that are sampled from a distribution

$$X_{i,j,k} \sim N(\mu_{i,j}, \sigma^2), k = 1, \dots, c$$

- Assume

$$\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij}$$

α_i: i-th row effect; β_j: j-th column effect; γ_{ij}: (i,j)-th interaction effect

Remark: without loss of generality, we assume $\sum_{i=1}^{a} \alpha_i = 0$ and $\sum_{j=1}^{b} \beta_j = 0$

$\sum_{i=1}^{a} \gamma_{ij} = 0$ and $\sum_{j=1}^{b} \gamma_{ij} = 0$.

Example

Factor 2
(b=4 levels)

	Brand of Gasoline							
Car	1		2		3		4	
1	31.0	24.9	26.3	30.0	25.8	29.4	27.8	27.3
	26.2	28.8	25.2	31.6	24.5	24.8	28.2	30.4
2	30.6	29.5	25.5	26.8	26.6	23.7	28.1	27.1
	30.8	28.9	27.4	29.4	28.2	26.1	31.5	29.1
3	24.2	23.1	27.4	28.1	25.2	26.7	26.3	26.4
	26.8	27.4	26.4	26.9	27.7	28.1	27.9	28.8

Factor 1
(a=3 levels)

Notations

Sample mean of the (i,j)-th interaction effect $\overline{X}_{ij\cdot} = \frac{1}{c}\sum_{k=1}^{c} X_{ijk},$

Sample mean of the i-th row factor $\overline{X}_{i\cdot\cdot} = \frac{1}{bc}\sum_{j=1}^{b}\sum_{k=1}^{c} X_{ijk},$

Sample mean of the j-th column factor $\overline{X}_{\cdot j\cdot} = \frac{1}{ac}\sum_{i=1}^{a}\sum_{k=1}^{c} X_{ijk},$

Overall sample mean $\overline{X}_{\cdots} = \frac{1}{abc}\sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{c} X_{ijk},$

Notations

- Overall sum-of-squares or total sum-of-squares

$$SS(TO) = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{c}\left(X_{ijk} - \overline{X}...\right)^2$$

- Sum-of-squares of Factor A (row factor)

$$SS(A) = bc\sum_{i=1}^{a}\left(\overline{X}_{i..} - \overline{X}...\right)^2$$

- Sum-of-squares of Factor B (column factor)

$$SS(B) = ac\sum_{j=1}^{b}\left(\overline{X}_{.j.} - \overline{X}...\right)^2$$

Notations

- Interaction sum-of-squares

$$SS(AB) = c\sum_{i=1}^{a}\sum_{j=1}^{b}(\overline{X}_{ij\cdot} - \overline{X}_{i\cdot\cdot} - \overline{X}_{\cdot j\cdot} + \overline{X}_{\cdot\cdot\cdot})^2$$

- Error sum-of-squares

$$SS(E) = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{c}\left(X_{ijk} - \overline{X}_{ij\cdot}\right)^2$$

- Relations

$$SS(TO) = SS(A) + SS(B) + SS(E) + SS(AB)$$

Two-factor Hypothesis Testing

We want to test:

$$\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij}$$

i-th row effect; j-th column effect; (i,j)-th interaction effect

1. (Is there an effect on μ due to row?)

$$H_{0A} : \alpha_1 = \alpha_2 = \ldots = \alpha_a = 0, \quad H_{1A} : H_{0A} \text{ is not true}$$

2. (Is there an effect on μ due to column?)

$$H_{0B} : \beta_1 = \beta_2 = \ldots = \beta_b = 0, \quad H_{1B} : H_{0B} \text{ is not true}$$

3. (Is there an effect on μ due to interactions between the row and the column?)

$$H_{0AB} : \gamma_{ij} = 0, \text{ for all } i, j, \quad H_{1AB} : H_{0AB} \text{ is not true}$$

Hypothesis Testing

$$H_{0A} : \alpha_1 = \alpha_2 = \ldots = \alpha_a = 0, \quad H_{1A} : H_{0A} \text{ is not true}$$

$$H_{0B} : \beta_1 = \beta_2 = \ldots = \beta_b = 0, \quad H_{1B} : H_{0B} \text{ is not true}$$

$$H_{0AB} : \gamma_{ij} = 0, \text{ for all } i, j, \quad H_{1AB} : H_{0AB} \text{ is not true}$$

If H_{0A}, H_{0B} and H_{0AB} are true, then

$$\underbrace{\frac{SS(TO)}{\sigma^2}}_{\sim\chi^2(abc-1)} = \underbrace{\frac{SS(A)}{\sigma^2}}_{\sim\chi^2(a-1)} + \underbrace{\frac{SS(B)}{\sigma^2}}_{\sim\chi^2(b-1)} + \underbrace{\frac{SS(E)}{\sigma^2}}_{\sim\chi^2(ab(c-1))} + \underbrace{\frac{SS(AB)}{\sigma^2}}_{\sim\chi^2((a-1)(b-1))},$$

Recall one-factor ANOVA

$$\mathrm{SS(TO)} = \mathrm{SS(E)} + \mathrm{SS(T)}.$$

$\sim \chi^2(n-1)$ $\sim \chi^2(m-1)$

Hypothesis Testing

Hypothesis

$H_{0A}: \alpha_1 = \alpha_2 = \ldots = \alpha_a = 0, \quad H_{1A}: H_{0A}$ is not true

Test statistics

$$F_A = \frac{\mathrm{SS(A)}/(a-1)}{\mathrm{SS(E)}/[ab(c-1)]}$$

Under H_{0A}:

$$F_A \sim F(a-1, ab(c-1))$$

Critical (rejection) region:

$$F_A \geq F_\alpha(a-1, ab(c-1))$$

***p*-value:**

$$P\{F(a-1, ab(c-1)) \geq F_A\}$$

Hypothesis

$H_{0B}: \beta_1 = \beta_2 = \ldots = \beta_b = 0, \quad H_{1B}: H_{0B}$ is not true

Test statistics

$$F_B = \frac{\mathrm{SS(B)}/(b-1)}{\mathrm{SS(E)}/[ab(c-1)]}$$

Under H_{0A}:

$$F_B \sim F(b-1, ab(c-1))$$

Critical (rejection) region:

$$F_B \geq F_\alpha(b-1, ab(c-1))$$

***p*-value:**

$$P\{F(b-1, ab(c-1)) \geq F_B\}$$

Hypothesis Testing

Hypothesis

$$H_{0AB}: \gamma_{ij} = 0, \text{ for all } i, j, \quad H_{1AB}: H_{0AB} \text{ is not true}$$

Test statistics

$$F_{AB} = \frac{\mathrm{SS(AB)}/[(a-1)(b-1)]}{\mathrm{SS(E)}/[ab(c-1)]}$$

Under H_{0A}:

$$F_{AB} \sim F\big((a-1)(b-1), ab(c-1)\big)$$

Critical (rejection) region:

$$F_{AB} \geq F_{\alpha}\big((a-1)(b-1), ab(c-1)\big)$$

***p*-value:**

$$P\big\{F\big((a-1)(b-1), ab(c-1)\big) \geq F_{AB}\big\}$$

Two-factor ANOVA Table

Table 9.4-4 Two-way ANOVA table, c observations per cell

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F
Factor A (row)	SS(A)	$a-1$	$\text{MS(A)} = \dfrac{\text{SS(A)}}{a-1}$	$\dfrac{\text{MS(A)}}{\text{MS(E)}}$
Factor B (column)	SS(B)	$b-1$	$\text{MS(B)} = \dfrac{\text{SS(B)}}{b-1}$	$\dfrac{\text{MS(B)}}{\text{MS(E)}}$
Factor AB (interaction)	SS(AB)	$(a-1)(b-1)$	$\text{MS(AB)} = \dfrac{\text{SS(AB)}}{(a-1)(b-1)}$	$\dfrac{\text{MS(AB)}}{\text{MS(E)}}$
Error	SS(E)	$ab(c-1)$	$\text{MS(E)} = \dfrac{\text{SS(E)}}{ab(c-1)}$	
Total	SS(TO)	$abc-1$		

Example

9.4-7. In order to test whether four brands of gasoline give equal performance in terms of mileage, each of three cars was driven with each of the four brands of gasoline. Then each of the (3)(4) = 12 possible combinations was repeated four times. The number of miles per gallon for each of the four repetitions in each cell is recorded in the following table:

	Brand of Gasoline							
Car	1		2		3		4	
1	31.0	24.9	26.3	30.0	25.8	29.4	27.8	27.3
	26.2	28.8	25.2	31.6	24.5	24.8	28.2	30.4
2	30.6	29.5	25.5	26.8	26.6	23.7	28.1	27.1
	30.8	28.9	27.4	29.4	28.2	26.1	31.5	29.1
3	24.2	23.1	27.4	28.1	25.2	26.7	26.3	26.4
	26.8	27.4	26.4	26.9	27.7	28.1	27.9	28.8

Example

Test the hypotheses H_{AB}: no interaction, H_A: no row effect, and H_B: no column effect, each at the 5% significance level.

Example:

$\bar{X}_{1..} = 27.6375$ $\bar{X}_{2..} = 28.08125$ $\bar{X}_{3..} = 26.7125$

$\bar{X}_{.1.} = 27.68333$ $\bar{X}_{.2.} = 27.58333$ $\bar{X}_{.3.} = 26.4$

$\bar{X}_{.4.} = 28.24167$

$\bar{X}_{...} = 27.47708$

$SS(TO) = 198.84$

$SS(A) = 15.61$. $SS(B) = 21.58$ $SS(AB) = 36.12$

$SS(E) = 125.53$

Example

ANOVA Table

Source	Sum of Squares (SS)	Degree of freedom	Mean Square	F	p-value
A	15.61	2	7.803	2.238	0.121
B	21.58	3	7.194	2.063	0.122
AB	36.12	6	6.021	1.727	0.143
Error	125.53	36	3.487		
Total	198.84	47			

Thresholds for H_A: $F_{0.05}(2, 36) = 3.259$

H_B: $F_{0.05}(3, 36) = 2.866$

H_{AB} $F_{0.05}(6, 36) = 2.364$

Fail to Reject H_A, H_B, H_{AB}

Summary

- **One-factor ANOVA**

Table 9.3-2 Analysis-of-variance table

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F Ratio
Treatment	SS(T)	$m-1$	$\text{MS(T)} = \frac{\text{SS(T)}}{m-1}$	$\frac{\text{MS(T)}}{\text{MS(E)}}$
Error	SS(E)	$n-m$	$\text{MS(E)} = \frac{\text{SS(E)}}{n-m}$	
Total	SS(TO)	$n-1$		

- **Two-factor ANOVA**

Table 9.4-1 Two-way ANOVA table, one observation per cell

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F
Factor A (row)	SS(A)	$a-1$	$\text{MS(A)} = \frac{\text{SS(A)}}{a-1}$	$\frac{\text{MS(A)}}{\text{MS(E)}}$
Factor B (column)	SS(B)	$b-1$	$\text{MS(B)} = \frac{\text{SS(B)}}{b-1}$	$\frac{\text{MS(B)}}{\text{MS(E)}}$
Error	SS(E)	$(a-1)(b-1)$	$\text{MS(E)} = \frac{\text{SS(E)}}{(a-1)(b-1)}$	
Total	SS(TO)	$ab-1$		

Table 9.4-4 Two-way ANOVA table, c observations per cell

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F
Factor A (row)	SS(A)	$a-1$	$\text{MS(A)} = \frac{\text{SS(A)}}{a-1}$	$\frac{\text{MS(A)}}{\text{MS(E)}}$
Factor B (column)	SS(B)	$b-1$	$\text{MS(B)} = \frac{\text{SS(B)}}{b-1}$	$\frac{\text{MS(B)}}{\text{MS(E)}}$
Factor AB (interaction)	SS(AB)	$(a-1)(b-1)$	$\text{MS(AB)} = \frac{\text{SS(AB)}}{(a-1)(b-1)}$	$\frac{\text{MS(AB)}}{\text{MS(E)}}$
Error	SS(E)	$ab(c-1)$	$\text{MS(E)} = \frac{\text{SS(E)}}{ab(c-1)}$	
Total	SS(TO)	$abc-1$		